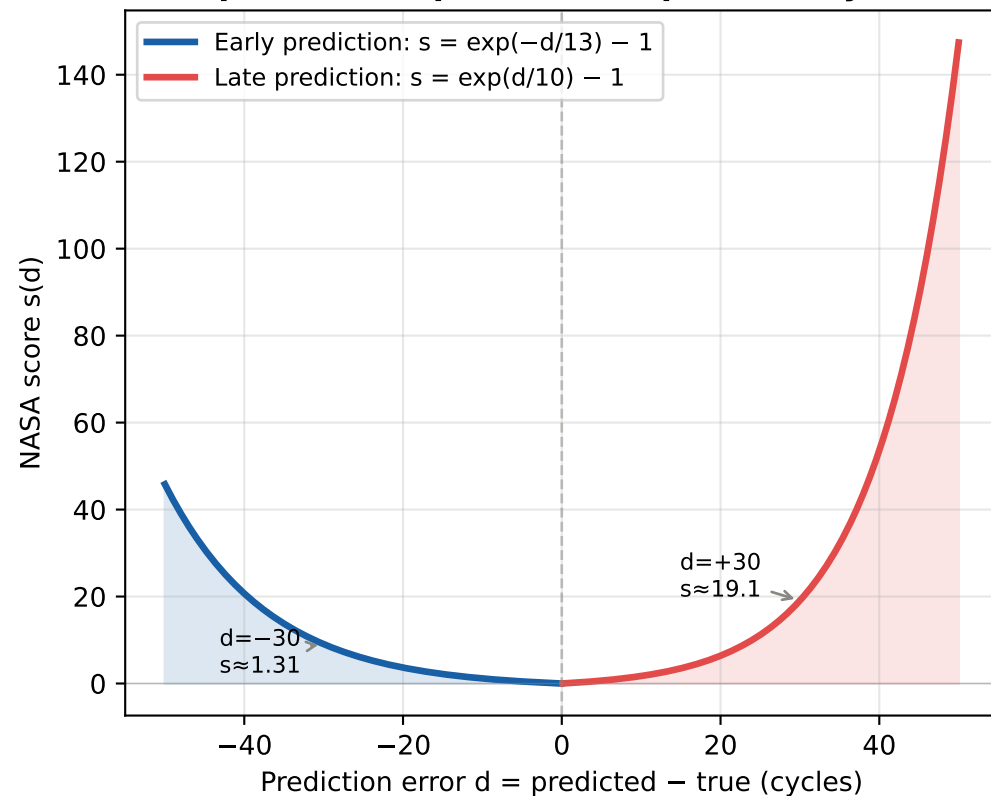


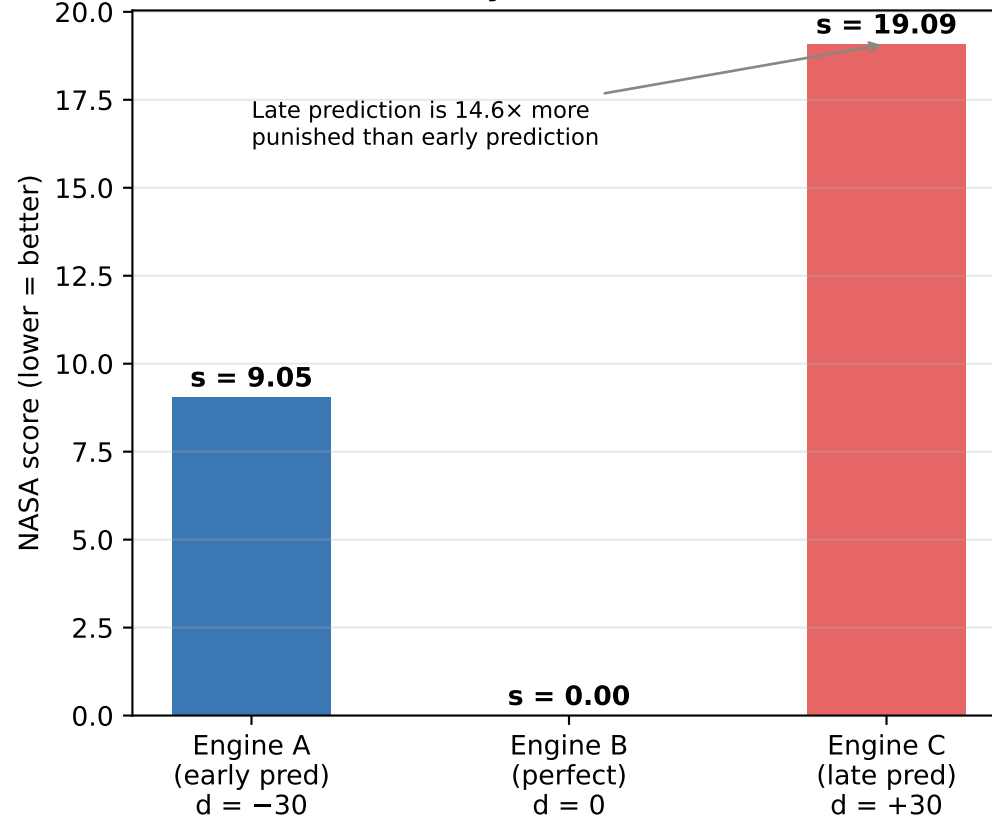
NASA Asymmetric Score — Definition, Examples, and MRO Interpretation

Penalises late predictions exponentially to reflect catastrophic failure risk

(a) NASA asymmetric penalty curve
Late predictions penalised exponentially harder



(b) Same absolute error ($|d|=30$) but asymmetric score



(c) Operational interpretation for MRO reviewers

NASA Score: operational interpretation

$d < 0$ (early prediction):

- Model says RUL = 200, true RUL = 230
- Engine inspected 30 cycles too early
- Cost: unnecessary maintenance
- Penalty: mild (exp scale factor 13)

$d > 0$ (late prediction):

- Model says RUL = 230, true RUL = 200
- Engine flies 30 cycles past safe limit
- Cost: potential catastrophic failure
- Penalty: severe (exp scale factor 10)

Why factor 10 vs 13?

The ratio $13/10 = 1.3$ encodes that in-flight failure is ~30% more catastrophic than unnecessary grounding.

ThermoPINN result:

NASA = 18.75 at 10-shot (lower = better) vs. LSTM baseline NASA = 78.5
→ 4.2× safer late-prediction behaviour